6.2 - Architectural Design

6.2.1 - Major Software Components

6.2.1.1 Data Contribution Mobile App

Application component dedicated to reporting microforest data from the field.
The Data Contribution CSC is responsible for the user interface and logic associated with field-side data entry. This component serves as the primary tool for volunteers and visitors to interact with and contribute to any of their local microforests. Its functionality includes managing user authentication through the Log In/Sign Up page, providing a structured Plant Data form with integrated QR scanning capabilities for automated ID entry, and maintaining a list of registered forests specific to the user. This section ensures that plant health logs are captured accurately and submitted to the central database in a timely and secure manner.

6.2.1.2 Administrator (Ranger) Portal

Application component dedicated to enabling admins to manage their microforest data.
The Administrator Portal CSC handles the administrative interface used by rangers and project managers. It provides a secure Sign In page to protect sensitive data and a comprehensive Dashboard page. The dashboard features an Analytics Window for headline statistics, a Users Window for contributor management, and a Data Window for granular logs review. It also includes a Forest Selector enabling the admin to select which forest they are managing and a "Download as CSV" functionality for local data processing and reporting.

6.2.1.3 Microforest Connect Web Interface

Visual web interface displaying microforest locations and environmental impact.
The Microforest Connect CSC represents the public-facing interactive web layer. It is responsible for compiling and displaying forest data to convey environmental impact. Key functions include a Map component that renders an interactive world map with "Microforest Pings" at specific geographic coordinates representing real microforests. When a user selects a forest, the Microforest Details page populates

with the project title, responsible stewards, and the physical scale of the site, among other information about its beneficial impact on the environment.

6.2.1.4 Backend Database

The data storage and manipulation system that serves as the project's central hub. The Backend Database CSC is the core storage and logic layer of the application. It is partitioned into two main units: the Forests Database CSU and the Users Database CSU. The Forests Database stores metadata, such as geographic coordinates, and maintains historical logs of plant growth and health metrics. The Users Database manages profile metadata and maintains the relational links between contributors and the forests they're registered to contribute to. It also handles security responsibilities, such as encrypting passwords and personal metadata.

6.2.2 - Major Software Interactions

6.2.2.1 - Data Contribution Mobile App and Backend Database

The Data Contribution Mobile App interacts with the Backend Database by submitting real-time plant health logs and user registration data. When a contributor submits a form, the app performs a write operation to the Forests Database, creating a historical log indexed by Plant ID. The database, in turn, provides the app with the list of "Registered Forests" associated with the authenticated user's profile.

6.2.2.2 - Administrator Portal and Backend Database

The Administrator Portal draws data from the Backend Database to populate the Analytics, Users, and Data windows on the dashboard. Changes made by an administrator, such as deleting erroneous reports or managing user accounts, are propagated back to the database to maintain system integrity. This interaction ensures that the granular plant logs stored in the Forests Database are accessible for CSV export and administrative review.

6.2.2.3 - Microforest Connect Web Interface and Backend Database

The Microforest Connect Web Interface acts as a read-only consumer of the Backend Database. It compiles forest metadata and calculated plant health statistics from the database to render map pings and fact sheets for the public. The database must ensure that the data provided to this interface is consistent with what is seen in the Admin Portal to maintain data integrity across the platform.

6.2.2.4 - Data Contribution Mobile App and Device Hardware

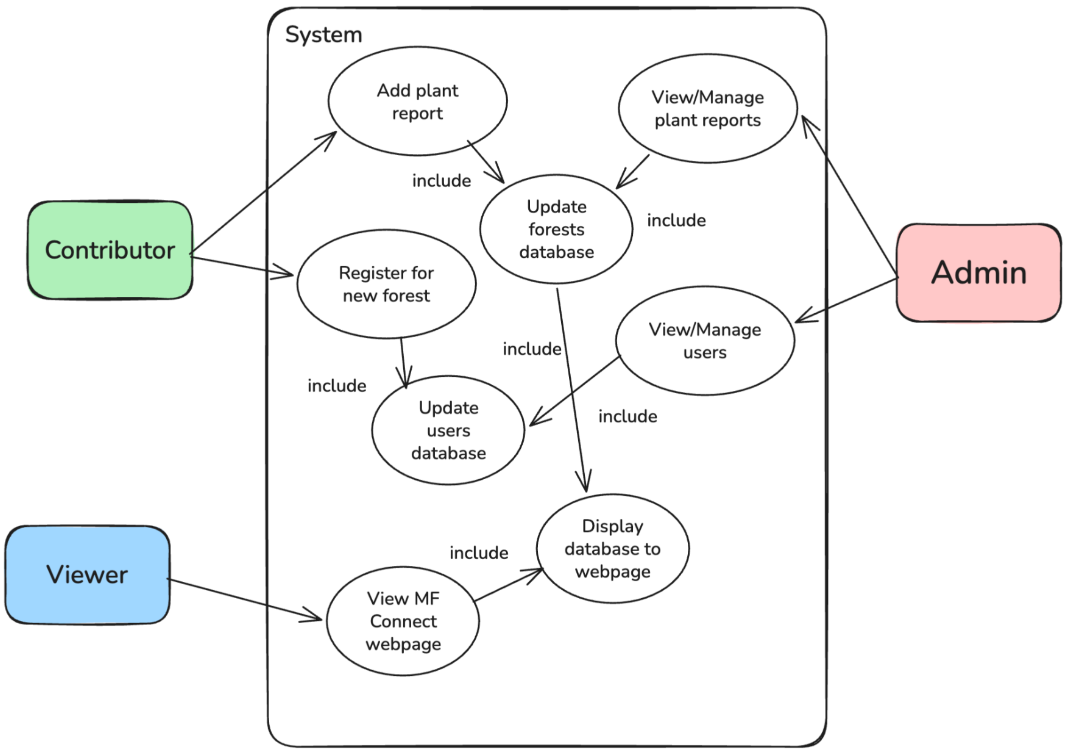
The Data Contribution App interfaces directly with the mobile device's hardware to fulfill its requirements. Specifically, it (optionally) utilizes the mobile camera to operate the QR Code Scanner for automated data entry. It also relies on high-brightness display capabilities and 4G/5G connectivity to facilitate real-time database synchronization while contributors are working in outdoor, possibly high-glare environments.

6.2.2.5 - Administrator Portal and Microforest Connect Interface

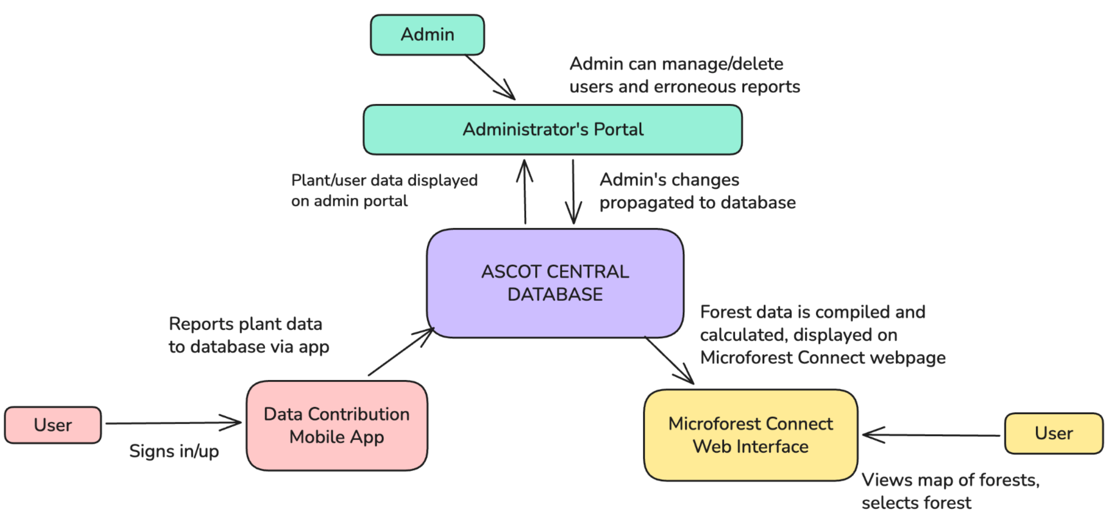
While these two components do not interact directly, they are synchronized through their shared reliance on the Backend Database. The Admin Portal manages and "cleans" the data that eventually becomes visible on the Microforest Connect Web Interface. This relationship ensures that only verified and managed data is presented to the public.

6.2.3 - Architectural Design Diagrams

Use Case Diagram



Component Diagram



Top-Down Class Diagram